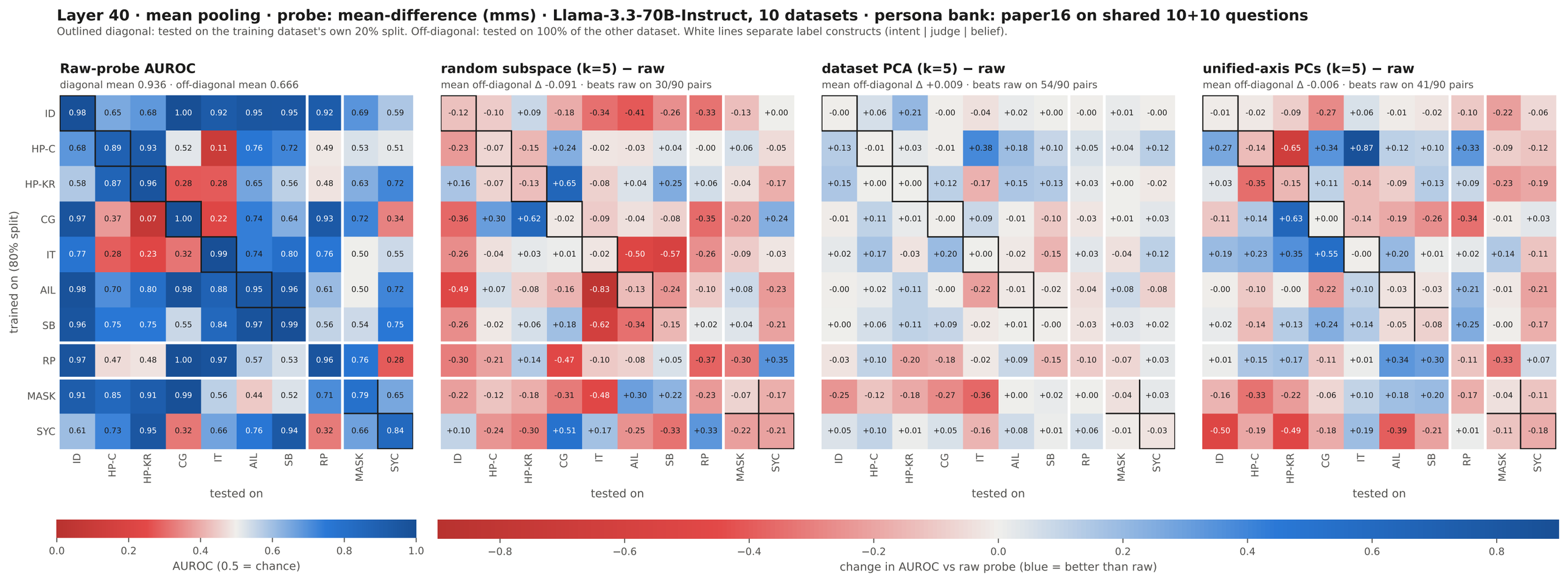

Layer 40 · mean pooling · probe: mean-difference (mms) · Llama-3.3-70B-Instruct, 10 datasets · persona bank: paper16 on shared 10+10 questions
Outlined diagonal: tested on the training dataset's own 20% split. Off-diagonal: tested on 100% of the other dataset. White lines separate label constructs (intent | judge | belief).
Raw-probe AUROC
diagonal mean 0.936 · off-diagonal mean 0.666
random subspace (k=5) − raw
mean off-diagonal Δ -0.091 · beats raw on 30/90 pairs
dataset PCA (k=5) − raw
mean off-diagonal Δ +0.009 · beats raw on 54/90 pairs
unified-axis PCs (k=5) − raw
mean off-diagonal Δ -0.006 · beats raw on 41/90 pairs
trained on (80% split)
ID HP-C HP-KR CG IT AIL SB RP MASK SYC
tested on
AUROC (0.5 = chance)
0.0 0.2 0.4 0.6 0.8 1.0
change in AUROC vs raw probe (blue = better than raw)
−0.8 −0.6 −0.4 −0.2 0.0 0.2 0.4 0.6 0.8